

Understanding Implied Information in Short English Sentences (G1)

A crucial aspect of human communication is the ability of a listener to **read between the lines**, grasping information that is not explicitly mentioned by the speaker but can still be inferred from their words. For instance, the sentence “Neil wants to host a dinner to say goodbye to his friends before he leaves” allows the reader to understand that Neil cares about his friends, that he will probably need to shop for ingredients, and that he is likely moving to another city, even though none of this is directly stated in the text.

The goal of this study is to better understand how artificial intelligence systems infer and represent such implicit meanings. To do this in a structured way, the system is given short snippets of narrative text and instructed to perform open information extraction, producing relational triplets of the form **(Subject, Relation, Object)**. These triplets represent both information that is explicitly mentioned in the text and information that the system derives from it. Subsequent steps are aimed at validating the inferred triplets, explaining how they were derived, and creating a timeline of the events described in the text.

Using the previous sentence as an example, some triplets that can be extracted are:

- (Neil, caresAbout, friends)
- (Neil, buys, ingredients)
- (Neil, wants, (Neil, hosts, dinner))

As you can see, a triplet can sometimes have another triplet as its object: this construct is used to represent subordination, such as in the snippet “Neil wants to host a dinner”.



1. Amazon MTurk Worker ID *

Inference classification

You will be asked questions about the quality of relational triplets extracted from a short piece of text.

Each triplet can either be explicitly stated in the text or only implicitly suggested by it. Sometimes the inferred triplets are actually wrong, because the information in the text is not enough to support them.

For each question, you will see:

- a short text (the snippet)
- a list of relational triplets extracted from that text

For each triplet, please indicate whether it is:

- **Factually expressed** in the text (explicitly stated)
- **Deducible / inferable** from the text (not stated, but clearly implied)
- **False or wrongly deduced** from the text (not supported or contradicted)

2. Context: **The man anticipated cold weather on his trip.** *

Factual

Deducible

False



(man,
anticipates,
(man,
experiences,
cold weather))

☐☐☐

(trip,
hasAttribute,
cold weather)

☐☐☐

(man,
experiences,
cold weather)

☐☐☐

(man,
anticipates, cold
weather)

☐☐☐

(man,
preparesFor,
trip)

☐☐☐

3. (trip, involves,
cold weather) **circus performer juggled while riding a unicycle. ***

Factual

Deducible

False

(circus
performer,
juggles,

☐☐☐

<none>)

(circus
performer,
rides, unicycle)

☐☐☐

(circus
performer,
performs,
juggling)

☐☐☐

(circus
performer,
isSkillfulAt,
juggling)

☐☐☐

(circus
performer,
isSkillfulAt,
riding)

☐☐☐

(juggling,
occursOn,
unicycle)

4.

ian pressed down on the moist flap of the envelope. *

Factual

Deducible

False

(man,
pressesDownOn
, flap)

☐☐☐

(flap, isPartOf, envelope)	☐	☐	☐
(flap, hasAttribute, moist)	☐	☐	☐
(man, interactsWith, flap)	☐	☐	☐
(flap, isMoist, <none>)	☐	☐	☐
(man, eats, envelope)	☐	☐	☐
(man, uses, envelope)	☐	☐	☐



5. Context: **The sale associate accused the girl of stealing.** *

	Factual	Deducible	False
(sale associate, accuses, girl)	☐	☐	☐
(girl, isAccusedOf, stealing)	☐	☐	☐
(sale associate, believes, girl)	☐	☐	☐
(sale associate, feels, (sale associate, isConcerned, <none>))	☐	☐	☐

6. (girl, steals, money) **ian received strange looks from passengers on the train.** *

	Factual	Deducible	False
(man, receives, (man, gets, strange looks))	☐	☐	☐



strangeLooks,,

(man, is,
subjectOfAttention)

☐☐☐

(man, gets,
strangeLooks)

☐☐☐

(passengers,
observe, man)

☐☐☐

(passengers,
feel, curiosity)

☐☐☐

(train, isEmpty,
<none>)

☐☐☐

(passengers,
express, strange
looks)

☐☐☐

(passengers,
areOn, train)

☐☐☐

(strangeLooks,
hasAttribute,
strange)

☐☐☐

Inference correction review

You will see a relational triplet that the AI model discarded because it considered it an unreasonable inference, together with the text snippet it was extracted from.

The AI is asked to follow three steps:

1. Decide whether the inferred triplet should be discarded.
2. Provide a reason for discarding it.
3. Propose a corrected triplet that fixes the issue (if possible).

You will be asked how much you agree with:

- the decision to discard the triplet,
- the reason given for discarding it,
- the corrected triplet proposed by the model.

As you progress through the questions, you will first see the discarded triplet, then the AI reasoning, and finally its proposed correction.



7. Context: **The man pressed down on the moist flap of the envelope.**

Discarded triplet: (man, uses, envelope)

Select how much you agree with the following statement:

The triplet should be discarded. *

- ☐ Fully agree
- ☐ Somewhat agree
- ☐ Disagree

8. Context: **The man pressed down on the moist flap of the envelope.**
Discarded triplet: (man, uses, envelope)
Reason for discarding: pressing down does not imply the man is using the envelope.

Select how much you agree with the following statement:

The reason for discarding the triplet is valid. *

- ☐ Fully agree
- ☐ Somewhat agree
- ☐ Disagree



9. Context: **The man pressed down on the moist flap of the envelope.**

Discarded triplet: (man, uses, envelope)

Reason for discarding: pressing down does not imply the man is using the envelope.

Corrected triplet: the model was unable to correct the triplet.

Select how much you agree with the following statement:

It's not possible to correct the triplet in a way that addresses the issue. *

- ☐ Fully agree
- ☐ Somewhat agree
- ☐ Disagree





10. Context: **The sale associate accused the girl of stealing.**

Discarded triplet: (sale associate, feels, (sale associate, isConcerned, <none>))

Select how much you agree with the following statement:

The triplet should be discarded. *

- ☐ Fully agree
- ☐ Somewhat agree
- ☐ Disagree



11. Context: **The sale associate accused the girl of stealing.**

Discarded triplet: (sale associate, feels, (sale associate, isConcerned, <none>))

Reason for discarding: the text does not indicate the sale associate's feelings or concerns.

Select how much you agree with the following statement:

The reason for discarding the triplet is valid. *

- ☐ Fully agree
- ☐ Somewhat agree
- ☐ Disagree

12. Context: **The sale associate accused the girl of stealing.**

Discarded triplet: (sale associate, feels, (sale associate, isConcerned, <none>))

Reason for discarding: the text does not indicate the sale associate's feelings or concerns.

Corrected triplet: the model was unable to correct the triplet.

Select how much you agree with the following statement:

It's not possible to correct the triplet in a way that addresses the issue. *

- ☐ Fully agree
- ☐ Somewhat agree
- ☐ Disagree

13. Context: **The man received strange looks from passengers on the train.**

Discarded triplet: (passengers, feel, curiosity)

Select how much you agree with the following statement:

The triplet should be discarded. *

- ☐ Fully agree
- ☐ Somewhat agree
- ☐ Disagree

14. Context: **The man received strange looks from passengers on the train.**

Discarded triplet: (passengers, feel, curiosity)

Reason for discarding: Strange looks do not necessarily indicate curiosity from the passengers.

Select how much you agree with the following statement:

The reason for discarding the triplet is valid. *

- ☐ Fully agree
- ☐ Somewhat agree
- ☐ Disagree

15. Context: **The man received strange looks from passengers on the train.**

Discarded triplet: (passengers, feel, curiosity)

Reason for discarding: Strange looks do not necessarily indicate curiosity from the passengers.

Corrected triplet: (passengers, express, strange looks)

Select how much you agree with the following statement:

The correction effectively addresses the issue. *

- ☐ Fully agree
- ☐ Somewhat agree
- ☐ Disagree

Event/State classification

For each question, you will be given a short text snippet and a list of relational triplets extracted from it. You will be asked to classify each triplet as either an **event** or a **state**.

In this setting, an event is a situation that happens or occurs; it can be instantaneous or last for a period of time. A state, on the other hand, is a condition or circumstance that holds true; it may remain constant throughout the text or change over time.

16. Context: **The man anticipated cold weather on his trip.** *

	Event	State
(man, anticipates, (man, experiences, cold weather))	☐	☐
(trip, hasAttribute,	☐	☐

	cold weather)	☐	☐
	(man, experiences, cold weather)	☐	☐
	(man, anticipates, cold weather)	☐	☐
	(man, preparesFor, trip)	☐	☐
17.	(trip, involves, cold weather)	rcus performer juggled while riding a unicycle. *	

	Event	State
(circus performer, juggles, <none>)	☐	☐
(circus performer, rides, unicycle)	☐	☐
(circus performer,	☐	☐

	performs, juggling)	☐	☐
	(circus performer, isSkillfulAt, juggling)	☐	☐
	(circus performer, isSkillfulAt, riding)	☐	☐
18.	(juggling, occursOn, unicycle)	ian pressed down on the moist flap of the envelope. *	

	Event	State
(man, pressesDownOn , flap)	☐	☐
(flap, isPartOf, envelope)	☐	☐
(flap, hasAttribute, moist)	☐	☐



(man,
interactsWith,
flap)

☐☐

(flap, isMoist,
<none>)

☐☐

(man, eats,
envelope)

☐☐

(man, uses,
envelope)

☐☐

19. Context: **The sale associate accused the girl of stealing.** *

	Event	State
(sale associate, accuses, girl)	☐	☐
(girl, isAccusedOf, stealing)	☐	☐
(sale associate, believes, girl)	☐	☐
(sale associate, feels, (sale associate, isConcerned, <none>))	☐	☐

20. (girl, steals, money) **ian received strange looks from passengers on the train.** *

	Event	State
(man, receives, (man, gets, strange looks))	☐	☐



strangeLooks,,

(man, is,
subjectOfAttention)

☐☐

(man, gets,
strangeLooks)

☐☐

(passengers,
observe, man)

☐☐

(passengers,
feel, curiosity)

☐☐

(train, isEmpty,
<none>)

☐☐

(passengers,
express, strange
looks)

☐☐

(passengers,
areOn, train)

☐☐

(strangeLooks,
hasAttribute,
strange)

☐☐

Event timing comparison

For each question, you will be given a snippet of text and two relational triplets extracted from it. Determine whether one of the triplets happens before the other, they happen simultaneously, or their time relation is vague.

21. Context: **The man anticipated cold weather on his trip.**

Triplet A: (man, anticipates, (man, experiences, cold weather))

Triplet B: (man, experiences, cold weather) *

- ☐ A occurs before B
- ☐ B occurs before A
- ☐ A and B occur simultaneously
- ☐ No clear relation

22. Context: **The circus performer juggled while riding a unicycle.**

Triplet A: (circus performer, juggles, <none>)

Triplet B: (circus performer, rides, unicycle) *

- ☐ A occurs before B
- ☐ B occurs before A
- ☐ A and B occur simultaneously
- ☐ No clear relation

23. Context: **The man pressed down on the moist flap of the envelope.**

Triplet A: (man, pressesDownOn, flap)

Triplet B: (man, interactsWith, flap) *

- ☐ A occurs before B
- ☐ B occurs before A
- ☐ A and B occur simultaneously
- ☐ No clear relation

24. Context: **The sale associate accused the girl of stealing.**

Triplet A: (sale associate, accuses, girl)

Triplet B: (girl, isAccusedOf, stealing) *

- ☐ A occurs before B
- ☐ B occurs before A
- ☐ A and B occur simultaneously
- ☐ No clear relation

25. Context: **The man received strange looks from passengers on the train.**

Triplet A: (man, receives, (man, gets, strangeLooks))

Triplet B: (passengers, express, strange looks) *

- ☐ A occurs before B
- ☐ B occurs before A
- ☐ A and B occur simultaneously
- ☐ No clear relation

Model error correction

As the model tries to infer all of the implicit information from the text, it might fail to capture some aspects of it. On the other hand, it could also make inferences that make little sense and fail to recognize its errors.

This is your chance to fix the model's mistakes and omissions. You will be given a context and all the relational triplets extracted from it. Based on these, you will be asked (i) to add any important triplets that you think are missing and (ii) to flag any triplets that you think should be discarded.

Regarding missing triplets, we invite you to add any important triplet you think are missing from the model's list. If you feel like nothing is missing, you can leave the textfield blank; otherwise, please report the triplets in the form (Subject, Relation, Object).

Remember that triplets can have <none> as the Object in case they represent a property, a state, or an event without a direct object.



26. Context: **The man anticipated cold weather on his trip.**

Model triplets:

(man, anticipates, (man, experiences, cold weather))

(trip, hasAttribute, cold weather)

(man, experiences, cold weather)

(man, anticipates, cold weather)

(man, preparesFor, trip)

(trip, involves, cold weather)

Important triplets that you feel are missing from the list above:



27. Are there any triplets you think should be **discarded**? Select ONLY the triplets that should be discarded (wrong / not supported). If none, select "None".

☐ (man, anticipates, (man, experiences, cold weather))

☐ (trip, hasAttribute, cold weather)

☐ (man, experiences, cold weather)

☐ (man, anticipates, cold weather)

☐ (man, preparesFor, trip)

☐ (trip, involves, cold weather)

☐ None

28. Context: **The circus performer juggled while riding a unicycle.**

Model triplets:

(circus performer, juggles, <none>)

(circus performer, rides, unicycle)

(circus performer, performs, juggling)

(circus performer, isSkillfulAt, juggling)

(circus performer, isSkillfulAt, riding)

(juggling, occursOn, unicycle)

Important triplets that you feel are missing from the list above:



29. Are there any triplets you think should be **discarded**? Select ONLY the triplets that should be discarded (wrong / not supported). If none, select "None".

- ☐ (circus performer, juggles, <none>)
- ☐ (circus performer, rides, unicycle)
- ☐ (circus performer, performs, juggling)
- ☐ (circus performer, isSkillfulAt, juggling)
- ☐ (circus performer, isSkillfulAt, riding)
- ☐ (juggling, occursOn, unicycle)
- ☐ None

30. Context: **The man pressed down on the moist flap of the envelope.**

Model triplets:

(man, pressesDownOn, flap)

(flap, isPartOf, envelope)

(flap, hasAttribute, moist)

(man, interactsWith, flap)

(flap, isMoist, <none>)

Important triplets that you feel are missing from the list above:



31. Are there any triplets you think should be **discarded**? Select ONLY the triplets that should be discarded (wrong / not supported). If none, select "None".

☐ (man, pressesDownOn, flap)

☐ (flap, isPartOf, envelope)

☐ (flap, hasAttribute, moist)

☐ (man, interactsWith, flap)

☐ (flap, isMoist, <none>)

☐ None

32. Context: **The sale associate accused the girl of stealing.**

Model triplets:

(sale associate, accuses, girl)

(girl, isAccusedOf, stealing)

(sale associate, believes, girl)

Important triplets that you feel are missing from the list above:



33. Are there any triplets you think should be **discarded**? Select ONLY the triplets that should be discarded (wrong / not supported). If none, select "None".

☐ (sale associate, accuses, girl)

☐ (girl, isAccusedOf, stealing)

☐ (sale associate, believes, girl)

☐ None

34. Context: **The man received strange looks from passengers on the train.**

Model triplets:

(man, receives, (man, gets, strangeLooks))

(passengers, areOn, train)

(strangeLooks, hasAttribute, strange)

(man, gets, strangeLooks)

(man, is, subjectOfAttention)

(passengers, observe, man)

(passengers, express, strange looks)

Important triplets that you feel are missing from the list above:

35. Are there any triplets you think should be **discarded**? Select ONLY the triplets that should be discarded (wrong / not supported). If none, select "None".

☐ (man, receives, (man, gets, strangeLooks))

☐ (passengers, areOn, train)

☐ (strangeLooks, hasAttribute, strange)

☐ (man, gets, strangeLooks)

☐ (man, is, subjectOfAttention)

☐ (passengers, observe, man)

☐ (passengers, express, strange looks)

☐ None